

Examples

Mudasir

This file demonstrates the features provided by the mudasir class. Most examples are AI generated as a simple demo of the visual style of the various commands.

Math Macro Examples

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$
$$\text{let } f(x) = \int_0^x f(t) dt = 1$$
$$\int_0^x f(t) dt \xrightarrow{\text{red arrow}} 0$$

Derivatives:

$$\frac{dy}{dx}, \quad \frac{de^{x^2}}{dx} = 2xe^{x^2}$$

Partial derivatives:

$$\frac{\partial u}{\partial x}, \quad \frac{\partial^2 u}{\partial x^2}$$

Integrals:

$$\int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}$$

Differentials (explicit d):

$$\int f(t) dt$$

Vector calculus / del operator:

$$\vec{\nabla} f, \quad \vec{\nabla} \cdot \vec{F}$$

Laplace transform:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st} dt$$

Inline code example: `disp('Hello, world!')`

Theorem-like Environments

Theorem 0.0.1. Let $x \in \mathbb{R}$. Then $x^2 \geq 0$.

Theorem. The square of any real number is nonnegative (unnumbered).

Lemma 0.0.2. If $a > 0$ and $b > 0$ then $ab > 0$.

Lemma. An unnumbered lemma example.

Proposition 0.0.3. Proposition: addition of reals is commutative.

Corollary 0.0.4. Corollary: immediate consequence of the proposition.

Definitions, Claims, and Examples

Definition 0.0.5. Definition of continuity at a point: A function f is continuous at $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

Definition. An unnumbered definition.

Claim 0.0.6. This is a sample claim with a short justification.

Claim. Unnumbered claim demonstration.

Example 0.1. Example: compute $1 + 1 = 2$.

Problems and Exercises

Problem 0.0.7. Compute the integral $\int_0^1 x^2 dx$.

★ **Problem 0.0.8.** A highlighted problem with a star symbol.

Exercise 0.2. Exercise: Show that the derivative of e^x is e^x .

Exercise. Unnumbered exercise: verify a trivial identity.

Notes, Remarks, Questions, Answers

Note. This is a side note (unnumbered by style).

Remark 0.0.9. Remark: pay attention to edge cases.

Remark. An unnumbered remark.

Question — What is the limit of $\sin x/x$ as $x \rightarrow 0$?

Question — An unnumbered question.

Answer — Answer: 1.

Answer — An unnumbered answer entry.

Reminder — Reminder: check assumptions in proofs.

Reminder — Unnumbered reminder.

WARNING — Warning: don't divide by zero.

Warning — Unnumbered warning.

Solution, Walkthrough, and Custom List

Problem 0.0.10. Prove that $2 + 2 = 4$.

Solution. Trivial by arithmetic. □

Walkthrough. This is a brief walkthrough for a sample solution.

- (a) First step.
- (b) Second step.
- (c) Third step.

Code (minted)

Block code example (from 'test.m'):

```
1 function z = f(t, y)
2     z = zeros(2,1);
3     z(1) = y(2);
4     z(2) = (-5*y(2)) + (-6*y(1));
5 end
6
7 tspan = [0 5];
8 y0 = [0, 1];           % initial condition y(0) = 0, dy/dt(0) = 1
9 [t, y] = ode45(@f, tspan, y0);
10
```

```
11 plot(t, y(:,1), '-o')
12 xlabel('Time t')
13 ylabel('y(t)')
14 title('Solution of ODE')
15 grid on
16 hold on
17 exact = -exp(-3 * t)+exp(-2 * t);
18 plot(t, exact, '-r')
```